

STROTHER FIELD, KS, USA

WMO: 724502

Lat: 37.168N

Lon: 97.037W

Elev: 351

StdP: 97.18

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 13932

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-9.8	-18.0	0.8	-10.3	-15.9	1.0	-8.4	13.2	6.4	12.0	7.1	3.7	0	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	37.9	23.6	36.3	23.8	34.3	23.6	25.8	33.1	25.3	32.6	24.8	31.8	6.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.9	19.6	29.9	23.1	18.6	28.8	22.6	18.1	28.4	82.2	33.1	79.2	32.4	76.7	31.9	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.8	9.3	-17.2	39.9	3.1	1.9	-19.4	41.3	-21.3	42.4	-23.0	43.5	-25.3	44.9		
			-17.7	27.0	3.0	0.7	-19.8	27.5	-21.6	27.9	-23.3	28.3	-25.5	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	1.8	3.9	9.3	14.5	19.8	25.0	27.6	26.9	22.7	15.6	8.9	2.8	(6)
(7)		DBStd	10.38	5.83	6.33	5.97	5.14	4.39	3.30	3.08	3.27	4.24	5.21	5.54	5.63	(7)
(8)		HDD10.0	873	259	186	84	18	1	0	0	0	12	83	231		(8)
(9)		HDD18.3	2262	512	403	283	135	37	1	0	11	114	284	482		(9)
(10)		CDD10.0	2689	6	16	63	153	306	451	546	523	381	187	51	7	(10)
(11)		CDD18.3	1036	0	0	4	19	84	202	287	265	142	31	2	0	(11)
(12)		CDH23.3	11347	1	2	37	184	737	2164	3482	3056	1396	269	18	0	(12)
(13)		CDH26.7	5376	0	0	5	39	242	981	1842	1582	608	74	2	0	(13)
(14)	Wind	WSAvg	4.4	4.3	4.6	5.1	5.3	4.5	4.6	4.1	3.7	3.9	4.3	4.5	4.3	(14)
(15)	Precipitation	PrecAvg	952	27	36	75	93	130	152	101	100	73	80	52	35	(15)
(16)		PrecMax	1369	61	91	188	235	335	385	208	182	160	252	154	91	(16)
(17)		PrecMin	661	0	0	13	16	56	22	17	7	22	7	0	5	(17)
(18)		PrecStd	187	20	26	47	49	64	88	44	47	42	59	48	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.3	22.8	26.9	30.3	33.6	37.8	40.2	40.2	37.8	31.6	25.8	20.0	(19)
(20)			MCWB	11.0	13.7	16.6	19.4	21.7	23.5	23.5	23.4	22.7	20.5	16.3	13.8	(20)
(21)		2%	DB	17.0	19.9	23.8	27.5	31.3	35.4	38.0	37.8	34.0	28.7	22.7	17.2	(21)
(22)			MCWB	10.0	12.1	15.6	18.2	22.0	23.7	23.9	23.5	22.8	19.2	15.4	11.8	(22)
(23)		5%	DB	14.0	17.0	21.4	25.6	29.2	33.4	36.3	36.2	32.4	26.4	20.3	13.9	(23)
(24)			MCWB	8.0	10.8	14.2	17.1	21.2	23.6	24.2	23.5	22.6	18.4	14.4	9.3	(24)
(25)		10%	DB	11.1	13.9	19.0	23.0	27.5	32.0	34.2	33.9	30.4	24.0	17.7	11.7	(25)
(26)			MCWB	6.2	8.8	13.1	16.4	20.9	23.3	24.2	23.6	21.8	17.8	12.8	7.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	15.8	19.5	21.1	24.6	26.5	27.0	26.7	25.2	23.1	19.0	16.6	(27)
(28)			MCDWB	16.6	19.5	24.0	27.4	30.4	33.9	33.7	33.7	32.5	27.9	22.8	18.9	(28)
(29)		2%	WB	10.6	13.9	17.5	19.9	23.6	25.6	25.9	25.7	24.1	21.8	16.8	12.8	(29)
(30)			MCDWB	15.3	18.2	21.3	25.0	28.8	32.5	33.3	33.0	30.9	25.8	20.6	15.1	(30)
(31)		5%	WB	8.3	11.8	15.9	18.9	22.6	24.8	25.4	25.0	23.4	20.5	15.3	10.1	(31)
(32)			MCDWB	13.3	15.9	20.0	23.6	27.5	31.2	32.9	32.3	29.9	24.2	18.9	13.5	(32)
(33)		10%	WB	6.5	9.1	13.9	17.6	21.6	24.0	24.8	24.2	22.7	18.8	13.7	7.7	(33)
(34)			MCDWB	10.9	13.9	18.4	22.3	26.2	29.9	32.2	31.1	28.8	22.9	17.4	11.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	11.9	12.4	12.2	11.3	11.1	11.3	11.7	12.0	12.2	12.0	10.8	(35)
(36)			MCDBR	16.6	16.2	15.5	14.7	12.9	12.7	13.7	14.5	13.4	14.1	14.5	14.7	(36)
(37)			MCWBR	10.4	10.2	9.1	8.2	6.0	4.8	4.2	4.4	5.1	7.0	8.7	9.9	(37)
(38)		5% WB	MCDWR	15.1	14.5	12.6	12.1	11.0	11.1	11.3	11.6	11.5	10.9	12.1	12.4	(38)
(39)			MCWBR	10.2	10.2	8.5	7.8	6.1	5.1	4.5	4.8	5.0	6.8	8.5	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.293	0.301	0.340	0.378	0.414	0.419	0.429	0.430	0.386	0.342	0.303	0.284	(40)	
(41)		taud	2.488	2.476	2.395	2.314	2.285	2.302	2.315	2.254	2.365	2.475	2.517	2.526	(41)	
(42)		Ebn at Noon	903	938	925	902	871	863	851	842	865	877	883	892	(42)	
(43)		Edh at Noon	80	92	110	127	133	130	128	133	113	92	78	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.56	3.26	4.25	5.32	5.92	6.76	6.70	6.02	5.02	3.73	2.75	2.18	(44)	
(45)		RadStd	0.26	0.48	0.28	0.36	0.55	0.54	0.40	0.35	0.39	0.44	0.30	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page